

Penalty view

$$C_k \rightarrow \lambda_k \rightarrow \Sigma$$

$$\lambda_k = \beta_k \quad (c = 0)$$

exact settings

$\beta_k = \text{const}$

Fixed penalty

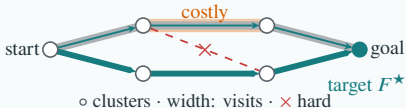
adaptive β_k

Lagrangian

SafeOT-Dual: one price rule

1 Budgeted flow

observed $\hat{F}$



2 price

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

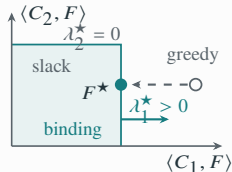
clipped c , filtered

feedback $\bar{J}_k - B_k$

3

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k} \rightarrow \text{PPO/TRPO}$$

Constraint view



$\varepsilon \rightarrow 0$: LP dual

clip saturates: active set

conditional limits